

RIVERSIDE HOSPITAL

WORKFLOW IMPLEMENTATION AND TESTING REPORT

Specialist Referral and Consultation Unit

Module	Development of Information Systems Project - UFCFAF-30-3
Scope	Strategic and operational workflow modelling, automation and testing
Technology	i*, BPMN, Camunda 8, Python, Git/GitHub
Testing environment	Camunda 8 local workflow environment, Modeler, Tasklist and Operate
Report version	Final evidence-aligned testing report
Report date	27 September 2026

Purpose

This report provides the submission-facing testing record for the Riverside Hospital workflow. It summarises the test strategy, current test scenarios, actual pathway coverage, evidence locations, resolved design issues, limitations and final readiness checks.

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1. Executive Summary

The Riverside Hospital project implements a coherent information-system prototype for a specialist referral and consultation unit. The workflow starts when a GP referral is received and ends when the final clinical outcome is communicated to the patient and referring GP. The implementation combines i* socio-technical modelling, strategic and operational BPMN, Camunda forms, process variables, Python service-task automation and Git/GitHub configuration management.

The assessment requires a focused testing strategy covering the process model, user forms, gateways, service tasks and key pathway scenarios. The final evidence structure therefore combines automated worker tests with end-to-end Tasklist and Operate execution evidence. This report has been updated to reflect the current TC01-TC10 evidence folders rather than the earlier four-test baseline in the original report.

Current test position

The current evidence set demonstrates service-task behaviour, an incomplete-referral clarification loop, laboratory-required pathways, no-laboratory discharge, follow-up outcome handling and treatment outcome handling. TC10 provides the clearest recent demonstration of the clarification-to-treatment route and reaches FINAL OUTCOME COMMUNICATED.

The current quality position is primarily evidence alignment rather than additional feature development. The repository and this report now use the same executable variable names, and the test matrix is aligned to the current TC01-TC10 evidence folders. TC02 remains partial evidence, while TC05 is now supported by a completed Operate instance showing the treatment pathway through final outcome communication. The assessment specification emphasises realistic scenarios, expected and actual results, defects and limitations, and clear evidence for service tasks, gateways and forms. This report is structured around those requirements.

2. Assessment Context and Project Objective

The assessment is a group-based development project in which groups of one are permitted. The project is expected to demonstrate software architecture analysis, information-system development, configuration management and software testing. Testing is explicitly required to cover the process model, user forms, gateways, service tasks and key pathway scenarios.

Assessment requirement	How the current project addresses it
Strategic Dependency / Rationale (i*)	Organisational actors, dependencies, goals, tasks, resources and softgoals were modelled for the Riverside referral process.
Strategic BPMN	A high-level AS-IS process captures the existing manual referral and consultation flow.
Operational BPMN	An executable TO-BE workflow uses human tasks, service tasks, gateways, forms, variables and message interaction.
Automation	Five automation roles were designed: referral completeness checking, appointment reference generation, laboratory appointment reference generation, results notification and draft outcome communication.
Testing	Current evidence covers worker-level automation and multiple end-to-end normal and exception routes.
Configuration management	Git/GitHub is used for models, forms, source code, documentation and testing evidence.

The assessment specification asks for a testing report showing scenarios, input data, expected results, actual results, pass/fail status, defects and limitations, with evidence that service tasks, gateways and forms have been tested. The current report follows that structure.

3. Case Study and Process Scope

The Riverside Hospital case study describes a specialist outpatient unit receiving referrals from independent GP practices. The prototype focuses on the administrative and clinical coordination from referral receipt to final outcome communication. Emergency care, inpatient treatment and real integrations with NHS, GP, laboratory, email or patient-notification systems remain outside the prototype scope.

Participant	Role in the workflow
GP Practice	External participant that sends the referral, supplies clarification when required and receives the final outcome.
Patient	External participant that receives appointment information and the final outcome communication.
Patient Booking Team	Captures or refines referral information; manages clarification; schedules initial, laboratory, follow-up and treatment appointments.
Specialist Doctor	Conducts consultation; records clinical information; determines laboratory need; reviews results; confirms final outcome.
Laboratory Team	Performs laboratory investigation and uploads results.
Medical Secretary	Reviews/finalises generated communications and confirms final communication.
Workflow Automation	Performs background checks, reference generation, results notification and draft communication generation.

The operational workflow contains three major decision points: referral completeness, laboratory requirement and final outcome type. The intended values are referralComplete = true/false, labRequired = true/false and finalOutcomeType = discharge/followUp/treatment.

4. Testing Strategy and Environment

Testing was designed to validate both isolated automation functions and integrated workflow behaviour. All test data is fictional. Evidence is captured from the user-facing Tasklist, the engine-side Operate view and, where the service task itself is the subject of the test, the Python worker output.

Environment item	Used in testing
Camunda	Camunda 8 local workflow environment
Modeler	Camunda Modeler 5.49.0
Operational tools	Tasklist and Operate
Automation	Python and Camunda Python SDK
Development environment	Visual Studio Code
Source control	Git / GitHub

Evidence principle

Tasklist evidence proves what a human user saw and submitted. Operate evidence proves how the process instance moved through the BPMN and what variables existed. Python worker evidence proves that an automated service task was picked up and what values were returned. Not every test needs all three sources; the strongest evidence set is the smallest set that directly proves the behaviour being claimed.

Testing objective

Confirm that the workflow behaves as designed, that gateways make the intended routing decisions, that forms collect the

required data, and that service-task outputs are available to downstream activities.

5. Current Test Matrix and Results

The matrix below consolidates the current test notes and the synchronised evidence folder. It supersedes the earlier four-test baseline in the source report. Where the retained evidence notes and screenshot set disagree, the report identifies the issue rather than silently replacing one with the other.

ID	Scenario	Input / route	Expected result	Actual result / evidence position	Status
TC01	Complete end-to-end laboratory + follow-up	Lab Test Patient; TEST654321; FBC; 25/09/2026 10:00	Complete operational route including laboratory, results review, follow-up and final communication	Integrated execution reaches final communication stage	PASS
TC02	Incomplete referral / clarification	Incomplete referral; missing information triggers clarification	Incomplete referral enters clarification before normal progression	Clarification stages are evidenced; complete loop is not independently proven in this folder	PARTIAL
TC03	Laboratory investigation pathway	Lab Test Patient; TEST654321; FBC; 25/09/2026 10:00	Lab appointment, investigation, result submission, notification/review before final outcome	Laboratory stages are evidenced and route continues to final outcome	PASS
TC04	Complete referral -> no laboratory -> discharge	Lab Test Patient; discharge outcome	Validation, consultation, no-lab branch, discharge and final communication	Current evidence filenames show a discharge route through final communication and completion	PASS
TC05	Treatment outcome pathway	Test Patient 5; treatment outcome; no laboratory investigation	Treatment selected, treatment appointment activities completed and final communication reached	Completed treatment-path execution includes treatment scheduling, appointment confirmation, patient notification, final outcome communication and completed Operate evidence.	PASS
TC06	Laboratory route -> result review -> follow-up	Laboratory-required referral; follow-up outcome	Lab appointment, investigation, results upload/review, follow-up appointment and communication	Evidence folder contains full route plus completed Operate evidence	PASS
TC07	Laboratory route -> result review -> discharge	Laboratory-required referral; discharge outcome	Lab route, result review, discharge and final communication	Evidence folder contains full route plus completed Operate evidence	PASS
TC08	Laboratory route -> follow-up outcome	Laboratory-required referral; follow-up outcome	Lab pathway followed by follow-up scheduling and outcome communication	Evidence folder contains the pathway and completed Operate evidence	PASS
TC09	No laboratory ->	John Smith / no-lab	Consultation, no-lab	Current execution	PASS

ID	Scenario	Input / route	Expected result	Actual result / evidence position	Status
	discharge	scenario; discharge	decision, discharge outcome, communication and completion	reaches FINAL OUTCOME COMMUNICATED and completed Operate instance	
TC10	Incomplete referral -> clarification -> treatment	Daniel Mwangi; contact clarification; treatment outcome	Missing information identified, clarification received, referral corrected, treatment scheduled and final communication completed	Current execution reaches FINAL OUTCOME COMMUNICATED and completed Operate instance	PASS

6. Detailed Pathway Coverage

6.1 Complete no-laboratory discharge

The no-laboratory route is demonstrated by the discharge-focused evidence sets. The key evidence shows the laboratory decision, final clinical outcome selection, outcome communication review, final communication and completed process instance. The route is important because it demonstrates gateway behaviour without entering the laboratory branch.

6.2 Incomplete referral and clarification

The clarification route is demonstrated by Tasklist stages for incomplete referral, clarification request, clarification receipt and referral update. TC10 additionally provides a fully completed recent execution: the missing contact information was identified, clarification was received, the referral was corrected, validation succeeded and the process continued into treatment.

6.3 Laboratory-required pathway

The laboratory evidence demonstrates the optional branch: laboratory appointment scheduling, appointment confirmation/reference generation, laboratory investigation, result upload, result review and continuation to a clinical outcome. This directly supports the case study requirement for an explicit laboratory branch and results-review loop.

6.4 Follow-up outcome

Follow-up is represented in the evidence folders by next-appointment scheduling, appointment reference/confirmation and progression to the common communication stage. The current evidence set contains multiple follow-up executions, providing broader route coverage than a single happy-path run.

6.5 Treatment outcome

Treatment is represented by a dedicated treatment outcome form, treatment appointment scheduling, next-appointment confirmation and final communication. TC10 is the strongest current treatment execution because it includes the clarification loop before treatment, treatment scheduling and a completed Operate instance reaching FINAL OUTCOME COMMUNICATED.

7. Service Tasks, Gateways and Forms

7.1 Service-task automation

Worker / service task	Purpose	Key outputs / evidence
check_referral_completeness.py	Checks required referral fields and identifies missing information.	referralComplete, missingReferralFields, referralCheckMessage
generate_appointment_reference.py	Creates a Riverside appointment reference and confirmation.	appointmentReference, appointmentConfirmation
generate_lab_appointment_reference.py	Creates a laboratory appointment reference and confirmation.	labAppointmentReference and confirmation output
notify_specialist_results_available.py	Checks laboratory result availability and produces a specialist notification.	results-available Boolean and notification text
generate_draft_outcome_letter.py	Builds draft patient/GP outcome communication from clinical data.	draft/final communication variables depending on current binding

The project therefore exceeds the assessment minimum of three automated service tasks. The evidence set also includes worker-side tests for the core automation behaviours described in the source testing report.

7.2 Gateway coverage

Gateway	Variable	Values	Purpose
REFERRAL COMPLETE?	referralComplete	true / false	Continue to booking or enter clarification loop.
LABORATORY TEST REQUIRED?	labRequired	true / false	Continue directly to final outcome or enter laboratory route.
FINAL OUTCOME TYPE?	finalOutcomeType	discharge / followUp / treatment	Select downstream clinical pathway before common outcome communication.

7.3 Form coverage

The form set mirrors the operational BPMN. Forms exist for referral entry/update, clarification, initial appointment scheduling, appointment confirmation, consultation completion, consultation outcome, laboratory appointment, laboratory confirmation, laboratory investigation, result upload/review, final clinical outcome, next appointment, communication review and final outcome confirmation.

8. Defects and Corrective Actions

The most useful defects found during integration were design/data-alignment issues rather than platform failures. Recording them demonstrates iterative engineering and provides the critical-evaluation evidence expected by the assessment.

Issue	Why it mattered	Corrective action / current position
labRequired data type mismatch	A multi-value control returned a collection while the gateway expected a Boolean.	Changed the laboratory decision control to a single Boolean checkbox.
Inconsistent form/worker variable names	Generated values could exist but be unavailable to downstream tasks if keys differed.	A process-wide naming standard was documented and mismatches were reviewed.
Final outcome fields all visible	The final outcome form could collect unrelated information and weaken data quality.	Conditional visibility was defined for follow-up requirements, treatment plan and discharge advice.
Next-appointment confirmation separation	Follow-up/treatment messages risked carrying laboratory-specific variables.	A dedicated next-appointment confirmation form was defined.
Draft vs final communication ambiguity	Generated text could be treated as final before secretary review.	Draft patient/GP letters were separated from final reviewed communication.
Evidence coverage initially too narrow	A single happy-path execution did not prove exception behaviour or all final outcomes.	Evidence plan expanded to incomplete referral, laboratory, follow-up, discharge and treatment routes.

Current consistency check

The source report notes that an uploaded code snapshot still contained some older output names. Before the final submission, BPMN, forms, process-scope-and-variables documentation and Python return dictionaries should be checked against the same final variable names.

9. Evidence Capture and Repository Structure

The assessment expects a repository evidence area and a test report. The current testing area is structured by TC01-TC10, with descriptive, ordered screenshot names. This makes the evidence readable without opening every file and supports a direct link between the report and the repository.

Evidence area	Current contents
TC01-TC10 folders	Ordered screenshot sets and scenario notes for each test case.
Tasklist evidence	Human-entered form values, gateway decisions, clarification, scheduling, consultation, laboratory, outcome and communication stages.
Operate evidence	Process history, variables and completed process instances where captured.
Automation evidence	Worker-side outputs for core service-task checks and generated communication/reference behaviour.
Root README / QA notes	Folder orientation and final quality checks.
Testing report	This PDF as the human-readable submission-facing report.

Evidence naming standard

Evidence filenames use a consistent TC##_NN_description structure. The order follows the process path rather than the screenshot capture time. Supplementary files are clearly separated with a 90-series suffix where required.

10. Critical Evaluation and Limitations

- The implementation is a local prototype; it does not integrate with real NHS, GP, laboratory, email or patient-notification systems.
- All patient and GP information used for testing is fictional.
- Appointment and notification workers simulate business-system behaviour rather than reserving real resources or delivering real messages.
- Generated communications are draft text and remain subject to clinical and administrative review.
- TC02 remains a partial evidence set in its own folder; the fully completed clarification route is demonstrated in TC10.
- TC05 contains complete treatment-path evidence, including a completed Operate instance showing the treatment route through final outcome communication.
- The BPMN, forms, process documentation and Python workers were checked against the final variable naming standard; no remaining executable variable-name mismatch was identified.

These limitations do not prevent the prototype from being evaluated as an executable workflow. Instead, they define the boundary between a demonstrable academic prototype and a production clinical system.

11. Final Submission Readiness

The current evidence set supports a broader testing story than the original four-test report. The key routes represented are service-task automation, incomplete referral clarification, laboratory-required processing, no-laboratory discharge, follow-up and treatment outcomes. TC02 remains partial in its own folder, while TC05 now has complete treatment-path evidence including a completed Operate instance; TC10 also provides a completed clarification-to-treatment execution. The assessment criteria also emphasise critical discussion of defects, limitations and quality awareness; these are explicitly documented.

Final check	Current position	Action before commit
Test evidence organisation	Ready	Keep TC01-TC10 folders and descriptive filenames.
Test report alignment	Final	Use this final evidence-aligned report in the submission package.
TC02 clarification evidence	Partial in TC02; fully exercised in TC10	Retain limitation note or reconcile TC02 notes before final commit.
TC04 scenario notes vs screenshots	Aligned	Keep the corrected no-laboratory discharge notes and evidence sequence.
TC05 treatment evidence	Complete treatment-path case supported by completed Operate evidence.	Retain TC05 as PASS and reference TC05_16_completed_operate.png in the evidence folder.
Variable consistency	Checked	Retain the final names consistently across BPMN, forms, documentation and Python workers.
Repository / commits	Ready for staged commits	Commit report and each evidence set using meaningful commit messages.
Presentation readiness	Strong basis	Be prepared to explain normal vs exception testing, key defects and limitations.

Submission-facing position

The testing report is now designed to support the evidence folder rather than duplicate it. The folder contains the screenshots; this report explains what each scenario proves, how the evidence relates to the workflow, what defects were found and what limitations remain.

Appendix A - Core Process Variables

Group	Preferred variables
Referral	patientName; nhsNumber; dateOfBirth; contactDetails; referringGP; gpPractice; referralReason; referralPriority
Validation	referralComplete; missingReferralFields
Clarification	clarificationRequest; clarificationReceived
Initial appointment	consultationDate; consultationTime; specialistDoctor; appointmentReference; appointmentConfirmation
Consultation	consultationCompleted; consultationNotes; consultationOutcomeSummary; labRequired; laboratoryTestType; laboratoryTestReason
Laboratory appointment	labAppointmentDate; labAppointmentTime; labAppointmentReference; labAppointmentConfirmation
Laboratory results	laboratoryInvestigationCompleted; laboratoryResultDate; labResult; labResultAvailable; specialistResultNotification; laboratoryReviewNotes
Final outcome	finalClinicalOutcomeSummary; finalOutcomeType; clinicalPlan; followUpRequirements; treatmentPlan; dischargeAdvice
Next appointment	nextAppointmentDate; nextAppointmentTime; nextAppointmentNotes; nextAppointmentReference; nextAppointmentConfirmation
Communication	draftPatientOutcomeLetter; draftGPOutcomeLetter; finalPatientOutcomeCommunication; finalGPOutcomeCommunication; outcomeCommunicationConfirmed

Appendix B - Evidence Folder Index

The repository evidence folder contains the following test-case groupings. Counts below are based on the synchronised evidence package used for this report.

Test case	Image evidence files	Notes file
TC01	10	test-notes.md
TC02	21	test-notes.md
TC03	8	test-notes.md
TC04	13	test-notes.md
TC05	16	test-notes.md
TC06	14	test-notes.md
TC07	14	test-notes.md
TC08	13	test-notes.md + README.txt
TC09	9	test-notes.md
TC10	11	test-notes.md

The evidence screenshots are deliberately kept in the evidence folder rather than embedded throughout this report so that the repository remains the primary source for execution proof. The report provides the narrative and indexing layer; the TC01-TC10 folders provide the direct execution evidence.